```
Last login: Wed Oct 21 09:07:09 on ttys000
elliottchalcraft@dhcp-10-29-143-115 ~ % eval "$(ssh-agent -s)"

Agent pid 74150
elliottchalcraft@dhcp-10-29-143-115 ~ % ssh-add -/.ssh/elliottq-21cm
ssh-add: invalid option -- /
usage: ssh-add [-AcDdKkLlqvXx] [-E fingerprint_hash] [-H hostkey_file]
               [-h destination_constraint] [-S provider] [-t life]
               [file ...]
       ssh-add -s pkcs11 [-Cv] [certificate ...]
       ssh-add -e pkcs11
       ssh-add -T pubkey ...
elliottchalcraft@dhcp-10-29-143-115 ~ % ssh-add ~/.ssh/elliottq-21cm
Identity added: /Users/elliottchalcraft/.ssh/elliottq-21cm (elliottchalcraft@dhcp-10-31-173-235.dyn.MIT.E
DU)
elliottchalcraft@dhcp-10-29-143-115 ~ % ssh -i ~/.ssh/elliottq-21cm -L 8080:localhost:8080 jlab@scotty.mi
t.edu
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-186-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Oct 21 01:17:18 PM UTC 2026

  System load:                 5.23
  Usage of /:                  7.2% of 914.66GB
  Memory usage:                2%
  Swap usage:                  0%
  Temperature:                 36.0 C
  Processes:                   261
  Users logged in:             0
  IPv4 address for enp133s0f1: 18.18.83.218
  IPv6 address for enp133s0f1: fdf2:2cd1:390:410f:b273:a786:cab0:b2a3
  IPv6 address for enp133s0f1: fdf2:2cd1:390:410f:7aeb:248b:a5aa:13a6
  IPv6 address for enp133s0f1: fdf2:2cd1:390:410f:73a:a89e:774:e726
  IPv6 address for enp133s0f1: fdf2:2cd1:390:410f:1248:bd6c:fd76:4de2
  IPv6 address for enp133s0f1: fdf2:2cd1:390:410f:e0fe:fc9:939e:5706
  IPv6 address for enp133s0f1: fdf2:2cd1:390:410f:73:7604:ddaf:b63
  IPv6 address for enp133s0f1: fdf2:2cd1:390:410f:5033:1b06:f5e9:2fef
  IPv6 address for enp133s0f1: fdf2:2cd1:390:410f:eef4:bbff:fedc:e619

  => There is 1 zombie process.

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

36 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

30 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


*** System restart required ***
Last login: Wed Oct 21 13:12:18 2026 from 10.29.143.115
(base) jlab@scotty:~$ ls
chrishc_margs_config  dsheen_config   old_srt_configs  srt-py
chrishc_margs_data    dsheen_scripts  Pictures         Templates
claire_peyton_data    help            Public           Videos
Desktop               lunar-eclipse   Pulse_hunting    wendy_jue
Documents             miniconda3      snap             willhath-milag-config
```

```
Downloads             Music           sophia_mingwen   WR66-config
(base) jlab@scotty:~$ ps aux | grep srt_runner
jlab      460973  0.0  0.0 189900 33168 pts/3    S+   Oct19   0:00 /home/jlab/miniconda3/envs/srt-ops/bin
/python3.11 /home/jlab/miniconda3/envs/srt-ops/bin/srt_runner.py --config_dir=WR66-config
jlab      460976 71.5  0.2 2970024 287672 pts/3  Sl+  Oct19 1774:14 /home/jlab/miniconda3/envs/srt-ops/bi
n/python3.11 /home/jlab/miniconda3/envs/srt-ops/bin/srt_runner.py --config_dir=WR66-config
jlab      460977  2.4  0.1 2676572 186344 pts/3  Sl+  Oct19  60:52 /home/jlab/miniconda3/envs/srt-ops/bin
/python3.11 /home/jlab/miniconda3/envs/srt-ops/bin/srt_runner.py --config_dir=WR66-config
jlab      461029  360  0.3 7005360 459320 pts/3  Sl+  Oct19 8932:15 /home/jlab/miniconda3/envs/srt-ops/bi
n/python3.11 /home/jlab/miniconda3/envs/srt-ops/bin/srt_runner.py --config_dir=WR66-config
jlab      468483  3.1  0.0      0     0 pts/3    Z+   14:00   0:06 [srt_runner.py] <defunct>
jlab      468502  0.0  0.0   6612  2328 pts/0    S+   14:04   0:00 grep --color=auto srt_runner
(base) jlab@scotty:~$ man screen
(base) jlab@scotty:~$ screen -list
There are screens on:
        428002.pts-0.scotty     (10/14/2026 06:36:31 PM)        (Detached)
        422778.pts-0.scotty     (10/13/2026 06:07:12 PM)        (Detached)
2 Sockets in /run/screen/S-jlab.
(base) jlab@scotty:~$ man screen
(base) jlab@scotty:~$ screen -r
There are several suitable screens on:
        428002.pts-0.scotty     (10/14/2026 06:36:30 PM)        (Detached)
        422778.pts-0.scotty     (10/13/2026 06:07:11 PM)        (Detached)
Type "screen [-d] -r [pid.]tty.host" to resume one of them.
(base) jlab@scotty:~$ screen -r 428002
[screen is terminating]
(base) jlab@scotty:~$ screen -list
There is a screen on:
        422778.pts-0.scotty     (10/13/2026 06:07:11 PM)        (Detached)
1 Socket in /run/screen/S-jlab.
(base) jlab@scotty:~$ screen -r
[screen is terminating]
(base) jlab@scotty:~$ screen -list
No Sockets found in /run/screen/S-jlab.

(base) jlab@scotty:~$ screen
[detached from 468562.pts-0.scotty]
(base) jlab@scotty:~$ screen -list
There is a screen on:
        468562.pts-0.scotty     (10/21/2026 02:13:43 PM)        (Detached)
1 Socket in /run/screen/S-jlab.
(base) jlab@scotty:~$ ps aux | grep srt_runner
jlab      468622  0.6  0.0 189900 33132 pts/1    S+   14:19   0:00 /home/jlab/miniconda3/envs/srt-ops/bin
/python3.11 /home/jlab/miniconda3/envs/srt-ops/bin/srt_runner.py --config_dir=WR66-config
jlab      468626 62.9  0.2 2891472 295356 pts/1  Sl+  14:20   0:39 /home/jlab/miniconda3/envs/srt-ops/bin
/python3.11 /home/jlab/miniconda3/envs/srt-ops/bin/srt_runner.py --config_dir=WR66-config
jlab      468627  6.3  0.1 2669428 154604 pts/1  Sl+  14:20   0:03 /home/jlab/miniconda3/envs/srt-ops/bin
/python3.11 /home/jlab/miniconda3/envs/srt-ops/bin/srt_runner.py --config_dir=WR66-config
jlab      468679  322  0.3 7006244 457252 pts/1  Sl+  14:20   3:00 /home/jlab/miniconda3/envs/srt-ops/bin
/python3.11 /home/jlab/miniconda3/envs/srt-ops/bin/srt_runner.py --config_dir=WR66-config
jlab      468767  0.0  0.0   6612  2312 pts/0    S+   14:21   0:00 grep --color=auto srt_runner
(base) jlab@scotty:~$ ls
chrishc_margs_config  dsheen_config   old_srt_configs  srt-py
chrishc_margs_data    dsheen_scripts  Pictures         Templates
claire_peyton_data    help            Public           Videos
Desktop               lunar-eclipse   Pulse_hunting    wendy_jue
Documents             miniconda3      snap             willhath-milag-config
Downloads             Music           sophia_mingwen   WR66-config
(base) jlab@scotty:~$ cd Documents
(base) jlab@scotty:~/Documents$ ls
(base) jlab@scotty:~/Documents$ cd ..
(base) jlab@scotty:~$ ls
chrishc_margs_config  dsheen_config   old_srt_configs  srt-py
chrishc_margs_data    dsheen_scripts  Pictures         Templates
claire_peyton_data    help            Public           Videos
Desktop               lunar-eclipse   Pulse_hunting    wendy_jue
Documents             miniconda3      snap             willhath-milag-config
Downloads             Music           sophia_mingwen   WR66-config
(base) jlab@scotty:~$ cd Downloads
```

```
(base) jlab@scotty:~/Downloads$ ls
 49_point_npoint_sinc_interpolated.png
 49_point_sunscan_countour_no_sinc_interpolation.png
 Andromeda_20240217.png
 CasA_1415_MHz_20240218.png
 Crab_Nebula_1415_MHz_20240218.png
 examplescriptrecording.fits
 g130_calon_3.18.2024.fits
 G140_caloff_3.18.2024.fits
 g140_calon_3.18.2024.fits
 G150_caloff_3.18.2024.fits
 g150_calon_3.18.2024.fits
 milkyway1.fits
 moon_1420_MHz_20240221.png
 moon_1420_MHz_morepoints_20240221.png
'newplot(1).png'
 newplot.png
 Orion_Nebula_1415_MHz_20240218.png
 Orion_Nebula_1420MHz_20240218.png
 SunScan_2024_02_12_0817.png
 Sunscan_20240220.png
(base) jlab@scotty:~/Downloads$ cd ..
(base) jlab@scotty:~$ cd Public
(base) jlab@scotty:~/Public$ ls
(base) jlab@scotty:~/Public$ cd ..
(base) jlab@scotty:~$ cd Desktop
(base) jlab@scotty:~/Desktop$ ls
(base) jlab@scotty:~/Desktop$ cd ..
(base) jlab@scotty:~$ ps aux | grep elliott_niko
jlab      468900  0.0  0.0   6612  2388 pts/0    S+   14:41   0:00 grep --color=auto elliott_niko
(base) jlab@scotty:~$ find . -type d -name "niko_elliott_data"
(base) jlab@scotty:~$ cd data
-bash: cd: data: No such file or directory
(base) jlab@scotty:~$ cd ..
(base) jlab@scotty:/home$ ls
akap  dsheen  jlab  nhaefner  olt  w1xm-admin
(base) jlab@scotty:/home$ cd jlab
(base) jlab@scotty:~$ ls
chrishc_margs_config  dsheen_config   old_srt_configs  srt-py
chrishc_margs_data    dsheen_scripts  Pictures         Templates
claire_peyton_data    help            Public           Videos
Desktop               lunar-eclipse   Pulse_hunting    wendy_jue
Documents             miniconda3      snap             willhath-milag-config
Downloads             Music           sophia_mingwen   WR66-config
(base) jlab@scotty:~$ cd ..
(base) jlab@scotty:/home$ cd ..
(base) jlab@scotty:/$ ls
bin   dev   lib    libx32      mnt   root  snap      sys  var
boot  etc   lib32  lost+found  opt   run   srv       tmp
data  home  lib64  media       proc  sbin  swap.img  usr
(base) jlab@scotty:/$ cd data
(base) jlab@scotty:/data$ ls
akap  davidct  dsheen  dsheen_temp  jlab  lost+found
(base) jlab@scotty:/data$ cd jlab
(base) jlab@scotty:/data/jlab$ ls
24-065-14_10_35.rad
24-065-14_16_24.rad
aaron_2024
aidan_teo_10-17
aidan_teo_data
aidan_teo_data_10_30
aidan_teo_data_floating_point_incorrect
alexgale
alex_jiwoo
andychen_sp25
az_scan_sun
bensam_data
cal_0_0off_1off_20260511.fits
cal_0_0off_1on_20260511.fits
```

cal_0_0on_1off_20260511.fits
cal_0_0on_1on_20260511.fits
cal_1_0off_1off_20260511.fits
cal_1_0off_1on_20260511.fits
cal_1_0on_1off_20260511.fits
cal_1_0on_1on_20260511.fits
cal_2_0off_1off_20260511.fits
cal_2_0off_1on_20260511.fits
cal_2_0on_1off_20260511.fits
cal_2_0on_1on_20260511.fits
cal_3_0off_1off_20260511.fits
cal_3_0off_1on_20260511.fits
cal_3_0on_1off_20260511.fits
cal_3_0on_1on_20260511.fits
cal_4_0off_1off_20260511.fits
cal_4_0off_1on_20260511.fits
cal_4_0on_1off_20260511.fits
cal_4_0on_1on_20260511.fits
cal_stars
claire_peyton_data
cold_sky.fits
cold_sky_plus_cal.fits
coupling_2024
crab_20250121_ringbuffer_test_2_16MHz
davidct
drf_ringbuffer_test_1_16MHz
eclipse_track_20240408
elevation_temperature_sweep_20250401-2.fits
elevation_temperature_sweep_20250401.fits
ellandy_data
historical_data
jbenke
katherine_tyra
kj
lily_liam
lunar_eclipse_03-13-2025
michael_nishant
niko_elliott_data
oris_data
pc_az_scan_sun.fits
phasecal
piero_margulan
pulsar_project_iap_2025
RotCurve_davidct
samimegan_data
Solar_davidct
sophia_mingwen
sophie_erin_data
sophie_erin_data_2
spatrick
SRT_RAW_SAVE-2025_09_30_04_33_13
SRT_RAW_SAVE-2025_10_15_14_47_39
SRT_SPEC_SAVE-2025_09_30_04_04_14.fits
SRT_SPEC_SAVE-2025_10_07_22_17_33.fits
SRT_SPEC_SAVE-2025_10_13_22_21_24.fits
SRT_SPEC_SAVE-2025_10_13_22_37_25.fits
SRT_SPEC_SAVE-2025_10_14_14_10_08.fits
SRT_SPEC_SAVE-2025_10_16_22_09_09.fits
SRT_SPEC_SAVE-2025_10_20_14_10_32.fits
SRT_SPEC_SAVE-2025_10_20_23_00_34.fits
SRT_SPEC_SAVE-2025_10_21_22_29_20.fits
SRT_SPEC_SAVE-2025_10_21_22_49_38.fits
SRT_SPEC_SAVE-2025_10_21_23_09_20.fits
SRT_SPEC_SAVE-2026_02_04_20_14_28.fits
SRT_SPEC_SAVE-2026_02_04_21_16_03.fits
SRT_SPEC_SAVE-2026_02_04_21_20_42.fits
SRT_SPEC_SAVE-2026_02_26_16_05_48.fits
SRT_SPEC_SAVE-2026_02_26_16_27_15.fits
SRT_SPEC_SAVE-2026_03_19_15_29_06.fits

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SRT_SPEC_SAVE-2026_09_30_15_28_14.fits
SRT_SPEC_SAVE-2026_09_30_20_21_45.fits
SRT_SPEC_SAVE-2026_09_30_20_44_13.fits
SRT_SPEC_SAVE-2026_10_19_15_17_54.fits
SRT_SPEC_SAVE-2026_10_19_19_36_50.fits
SRT_SPEC_SAVE-2026_10_19_19_42_22.fits
SRT_SPEC_SAVE-2026_10_21_14_43_49.fits
test.fits
truji
wendy_jue
willhath-milag_data
(base) jlab@scotty:/data/jlab$ scp -r jlab@scotty.mit.edu:/data/jlab/SRT_SPEC_SAVE-2026_10_21_14_43_49.fi
ts /Users/elliottchalcraft/Documents/2026-27/8.13/jlab/21cm_radio/data/raw/10-21-2026/sun
jlab@scotty.mit.edu: Permission denied (publickey).
(base) jlab@scotty:/data/jlab$ scp -r jlab@scotty.mit.edu:/data/jlab/SRT_SPEC_SAVE-2026_10_21_14_43_49.fi
ts /Users/elliottchalcraft/Documents/2026-27/8.13/jlab/21cm_radio/data/raw/10-21-2026/sun/
jlab@scotty.mit.edu: Permission denied (publickey).
(base) jlab@scotty:/data/jlab$ mv SRT_SPEC_SAVE-2026_10_21_14_43_49.fits niko_elliott_data
(base) jlab@scotty:/data/jlab$ ls
24-065-14_10_35.rad
24-065-14_16_24.rad
aaron_2024
aidan_teo_10-17
aidan_teo_data
aidan_teo_data_10_30
aidan_teo_data_floating_point_incorrect
alexgale
alex_jiwoo
andychen_sp25
az_scan_sun
bensam_data
cal_0_0off_1off_20260511.fits
cal_0_0off_1on_20260511.fits
cal_0_0on_1off_20260511.fits
cal_0_0on_1on_20260511.fits
cal_1_0off_1off_20260511.fits
cal_1_0off_1on_20260511.fits
cal_1_0on_1off_20260511.fits
cal_1_0on_1on_20260511.fits
cal_2_0off_1off_20260511.fits
cal_2_0off_1on_20260511.fits
cal_2_0on_1off_20260511.fits
cal_2_0on_1on_20260511.fits
cal_3_0off_1off_20260511.fits
cal_3_0off_1on_20260511.fits
cal_3_0on_1off_20260511.fits
cal_3_0on_1on_20260511.fits
cal_4_0off_1off_20260511.fits
cal_4_0off_1on_20260511.fits
cal_4_0on_1off_20260511.fits
cal_4_0on_1on_20260511.fits
cal_stars
claire_peyton_data
cold_sky.fits
cold_sky_plus_cal.fits
coupling_2024
crab_20250121_ringbuffer_test_2_16MHz
davidct
drf_ringbuffer_test_1_16MHz
eclipse_track_20240408
elevation_temperature_sweep_20250401-2.fits
elevation_temperature_sweep_20250401.fits
ellandy_data
historical_data
jbenke
katherine_tyra
kj
lily_liam
lunar_eclipse_03-13-2025
```

```
michael_nishant
niko_elliott_data
oris_data
pc_az_scan_sun.fits
phasecal
piero_margulan
pulsar_project_iap_2025
RotCurve_davidct
samimegan_data
Solar_davidct
sophia_mingwen
sophie_erin_data
sophie_erin_data_2
spatrick
SRT_RAW_SAVE-2025_09_30_04_33_13
SRT_RAW_SAVE-2025_10_15_14_47_39
SRT_SPEC_SAVE-2025_09_30_04_04_14.fits
SRT_SPEC_SAVE-2025_10_07_22_17_33.fits
SRT_SPEC_SAVE-2025_10_13_22_21_24.fits
SRT_SPEC_SAVE-2025_10_13_22_37_25.fits
SRT_SPEC_SAVE-2025_10_14_14_10_08.fits
SRT_SPEC_SAVE-2025_10_16_22_09_09.fits
SRT_SPEC_SAVE-2025_10_20_14_10_32.fits
SRT_SPEC_SAVE-2025_10_20_23_00_34.fits
SRT_SPEC_SAVE-2025_10_21_22_29_20.fits
SRT_SPEC_SAVE-2025_10_21_22_49_38.fits
SRT_SPEC_SAVE-2025_10_21_23_09_20.fits
SRT_SPEC_SAVE-2026_02_04_20_14_28.fits
SRT_SPEC_SAVE-2026_02_04_21_16_03.fits
SRT_SPEC_SAVE-2026_02_04_21_20_42.fits
SRT_SPEC_SAVE-2026_02_26_16_05_48.fits
SRT_SPEC_SAVE-2026_02_26_16_27_15.fits
SRT_SPEC_SAVE-2026_03_19_15_29_06.fits
SRT_SPEC_SAVE-2026_09_30_15_28_14.fits
SRT_SPEC_SAVE-2026_09_30_20_21_45.fits
SRT_SPEC_SAVE-2026_09_30_20_44_13.fits
SRT_SPEC_SAVE-2026_10_19_15_17_54.fits
SRT_SPEC_SAVE-2026_10_19_19_36_50.fits
SRT_SPEC_SAVE-2026_10_19_19_42_22.fits
test.fits
truji
wendy_jue
willhath-milag_data
(base) jlab@scotty:/data/jlab$ cd niko_elliott_data
(base) jlab@scotty:/data/jlab/niko_elliott_data$ ls
SRT_SPEC_SAVE-2026_10_19_14_05_28.fits  SRT_SPEC_SAVE-2026_10_21_14_43_49.fits
(base) jlab@scotty:/data/jlab/niko_elliott_data$ screen -list
There is a screen on:
        468562.pts-0.scotty     (10/21/2026 02:13:42 PM)        (Detached)
1 Socket in /run/screen/S-jlab.
(base) jlab@scotty:/data/jlab/niko_elliott_data$ screen -r
[screen is terminating]
(base) jlab@scotty:/data/jlab/niko_elliott_data$ screen -list
No Sockets found in /run/screen/S-jlab.

(base) jlab@scotty:/data/jlab/niko_elliott_data$
```